Process Characterization Master Plan

A-Mab Drug Substance — PCMP-001

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
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Abbreviations

2-AB, 2-aminobenzamide; A280, ultraviolet absorbance at 280 nm; AEX, anion exchange chromatography; AMV, analytical method validation; BLA, Biologics License Application; CEX, cation exchange chromatography; CPP, critical process parameter; Cpk, process capability index; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; NTU, nephelometric turbidity unit; PAR, proven acceptable range; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; RPN, risk priority number; RSM, response surface methodology; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; TCID50, fifty per cent tissue culture infectious dose; UF/DF, ultrafiltration and diafiltration; UPLC, ultra performance liquid chromatography; UV, ultraviolet; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This master plan defines the scope, the strategy and the common methods of the Stage 1 process characterization campaign for A-Mab drug substance. The campaign covers the 8 unit operations of the drug substance process, from the production bioreactor through to ultrafiltration and diafiltration. Each unit operation is planned in its own protocol and reported in its own report (PCP-003 to PCP-010 and PCR-003 to PCR-010), and this plan states what those documents have in common, so that each protocol needs to describe only what is specific to its own step.

Four elements are common to every study in the campaign and are fixed here. The first is the register of quality attributes the process must control, with the acceptance criteria against which every result is judged (Section 3), and the second is the risk basis that decides both how each parameter is studied and how wide a range it is studied over (Section 4). The third is the approach to scale-down models and their qualification (Section 5). The fourth is the statistical method and the acceptance criteria framework that each protocol instantiates (Section 6 and Section 7).

The process train and the principal role of each step are shown in Figure 1.1 and listed in Table 1.1. The sequence is the platform purification train used for the sponsor's earlier humanized IgG1 products (Protein A capture, low pH hold, two polishing steps, virus filtration and formulation), so the mechanism of each step is established before the campaign begins and the studies confirm and bound it.

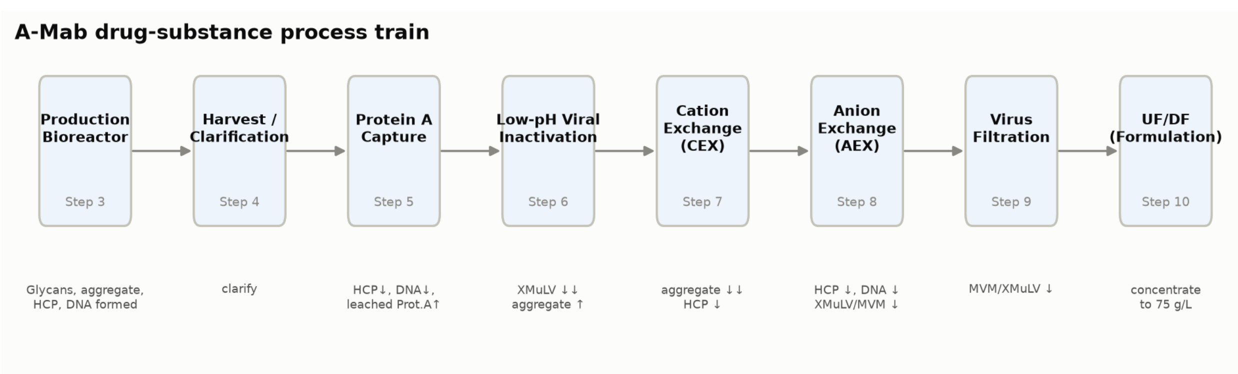

Figure 1.1: The A-Mab drug-substance process train, with the principal quality impact of each unit operation.

Table 1.1: The A-Mab drug-substance process train and the principal role of each unit operation.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

This plan sits under the process transfer plan (PTP-001), which defines the transfer of the drug substance process from the development site to the commercial facility. The pre-characterization risk assessment (RA-001) sets the scope of the studies. The individual protocols plan them, the individual reports document them, and the master report (PCMR-001) consolidates the campaign into one control strategy. The documents of the package are listed in Table 1.2.

Table 1.2: The A-Mab drug-substance document package, excluding this plan.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)

Document ID	Document class	Subject
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

Several activities lie outside this campaign. Stage 2 process performance qualification is planned separately, and the 5 qualification batches are not covered here. The seed expansion steps upstream of the production bioreactor (vial thaw through the N-1 seed bioreactor) are not characterized, because prior knowledge of the platform shows that they do not affect product quality (CMC Biotech Working Group 2009). Formulation and drug product operations are reported with the drug product, so ultrafiltration and diafiltration is included only for its effect on the delivered concentration and on mass balance. Validation of the analytical methods is performed under the method validation documents listed in Table 5.1, and it is not repeated in the characterization reports, which cite the validation and report only results obtained with the validated method.

The campaign does not on its own establish the viral safety of the process. Three steps (low pH inactivation, anion exchange and virus filtration) contribute clearance by independent mechanisms, their contributions are quantified in the reports for those steps, and the cumulative claim is made once in PCMR-001 (International Council for Harmonisation 2023a).

2 Stage 1 characterization strategy

Process characterization is the set of studies that link the operating parameters of each unit operation to the quality of the drug substance. It is the process design stage of the validation lifecycle (U.S. Food and Drug Administration 2011), and it produces the process understanding on which the commercial control strategy rests (International Council for Harmonisation 2009, 2012). The campaign is run in the enhanced approach, in which prior knowledge and a documented risk assessment decide what is studied, designed experiments quantify the effects of the parameters selected, and the outcome is a control strategy justified by data rather than by convention.

A-Mab is produced on an established humanized IgG1 platform. The purification sequence, the resin chemistries and the operating principles are those used for three earlier licensed products of the same class, and the manufacturing experience with that platform shows that the sequence is robust and delivers drug substance of consistent quality (CMC Biotech Working Group 2009). Prior knowledge is used in two ways here: it justifies the ranges taken into study, and it justifies studying univariately those parameters whose behaviour the platform already covers (Section 4). It does not replace step-specific data for any parameter linked to a critical quality attribute.

Three regions are distinguished throughout the campaign, and the terms are used the same way in every document. The knowledge space is the region over which the process has been characterized, and the design space is the subset of that region in which every critical quality attribute is known to be acceptable at the same time (International Council for Harmonisation 2009). The control space is the narrower region in which the process is routinely operated, and it is defined by the normal operating ranges (NORs) given in each plan.

The campaign covers 37 process parameters across the 8 unit operations. 22 of them are studied in multivariate designed experiments and 15 are studied univariately (Section 4). The designed experiments comprise 236 scale-down runs in total, of which 94 are screening runs and 142 belong to the response surface designs.

Every study in the campaign has the same objectives. It qualifies the scale-down model that generates its data, quantifies the effect of the step's parameters on the attributes the step governs, and defines the ranges over which the step can be operated (and, where the data support one, a design space). It also provides the evidence for the classification of each parameter and the capability estimate that the Stage 2 programme depends on.

Characterization does not confirm the process at commercial scale. Every study is executed at scale-down, its conclusions hold at manufacturing scale only so far as the scale-down model is qualified for the attributes concerned (Section 5), and the edges of an operating region are never run at commercial scale, so a design space is a modelled region and not a demonstrated one (CMC Biotech Working Group 2009). Confirmation at scale is the purpose of the Stage 2 qualification batches.

The campaign ends with a control strategy. Each parameter is classified from its study result and from the control capability available at the receiving site, each attribute is assigned to the steps that govern it, and the ranges that follow are the ones proposed for commercial operation (International Council for Harmonisation 2008, 2012). That consolidation is done once, in PCMR-001, because several attributes are governed by more than one step and no single report can settle them.

3 Critical quality attribute framework

The quality attributes in scope for the campaign are listed in Table 3.1 with their acceptance criteria and their assigned criticality. 10 attributes are in scope across 5 categories, and 6 of them are of high or very high criticality.

Table 3.1: Quality attributes in scope for the characterization campaign, with the step at which each is set, its acceptance criterion, its criticality level and its Tool #1 criticality score.

CQA	Set by	Category	Acceptance	Criticality	Tool #1
Afucosylation	Production Bioreactor	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	Production Bioreactor	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	Production Bioreactor	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	Production Bioreactor	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	Production Bioreactor	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	Production Bioreactor	process impurity	0–100 ng/mg	M-H	36
Residual DNA	Production Bioreactor	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	Protein A Chromatography	process impurity	0–5 ppm	L-M	16
Viral clearance — XMuLV (enveloped)	Low-pH Viral Inactivation	viral safety	16.7–30 log10 (cumulative)	VH	20
Viral clearance — MVM (parvovirus)	Anion Exchange Chromatography	viral safety	8.6–20 log10 (cumulative)	VH	20

Criticality is treated as a continuum and not as a binary label (International Council for Harmonisation 2023b). Each attribute carries a Tool #1 score, which is the product of its impact on safety or efficacy and the uncertainty in that impact, together with a severity score from the second assessment tool (CMC Biotech Working Group 2009). Attributes of high and very high criticality are treated as critical for the purposes of this campaign. The assessment itself is not repeated here: it is taken from the risk assessment (RA-001), and it fixes both which attributes each study is required to measure and which of them can constrain an operating range.

Three entries in the register drive the design of the campaign. High Mannose carries the highest criticality score of the register, is formed in the production bioreactor and has an acceptance range of 3–10 % of glycans. The two viral clearance attributes are cumulative over the whole train and are expressed as lower limits, at 16.7–30 log10 (cumulative) for the enveloped model virus (XMuLV) and 8.6–20 log10 (cumulative) for the parvovirus (MVM). Aggregates are formed upstream, raised by the low pH hold (Step 6) and reduced principally at cation exchange (Step 7), so their limit is met by a sequence of steps and not by any one of them.

7 of the 10 attributes are set in the production bioreactor, which is why that step carries the largest study of the campaign. Leached Protein A arises at capture and is cleared by the polishing steps that follow it. Host cell protein and residual DNA are generated upstream and are cleared by three chromatography steps in sequence. No single plan can therefore fix the acceptable range of a parameter that acts on host cell protein, because the acceptable output of one step depends on the operating conditions of the steps after it (CMC Biotech Working Group 2009), so each report gives its own step's contribution and names the reports that cover the others.

The acceptance criteria in Table 3.1 are drug substance limits and they are inputs to every study in the campaign. Several are set from clinical and manufacturing experience with the earlier products of the platform (the acceptance range for aggregates is one of them), and none of them is derived from the characterization data. Where a study needs an in-process limit for an intermediate pool, that limit is stated in the plan for the step and is justified against the drug substance limit it protects.

The register covers attributes of the drug substance. Attributes controlled by the drug product process are outside the campaign, and the studies that support them are reported with the drug product.

4 Risk-based prioritization

The risk assessment decides how each parameter is studied, and its outcome across the train is summarized in Table 4.1. RA-001 assesses every parameter of every step against the quality attributes it can affect. Of the 37 parameters in the campaign, 21 are linked to at least one quality attribute, 22 are carried into a multivariate designed experiment and 15 are studied univariately.

Table 4.1: Study scope handed to the characterization plans by RA-001, by unit operation. A parameter is counted as linked to a CQA where the risk assessment identifies at least one quality attribute at risk from it.

Step	Unit operation	Parameters	Linked to a CQA	Multivariate	Univariate	High priority
3	Production Bioreactor	9	5	5	4	5
4	Harvest and Clarification	3	0	0	3	0
5	Protein A Chromatography	6	2	4	2	0
6	Low-pH Viral Inactivation	4	3	3	1	3
7	Cation Exchange Chromatography	5	4	4	1	3
8	Anion Exchange Chromatography	5	5	4	1	3
9	Small-Virus Retentive Filtration	2	2	2	0	2
10	Ultrafiltration / Diafiltration	3	0	0	3	0

The ranking method is described in RA-001 and is not repeated here. Each parameter carries a severity score taken from the most severe attribute it can affect, an occurrence score for an excursion outside its normal operating range, and a detection score for the chance that such an excursion escapes routine monitoring, and the product of the three is the initial risk priority number (RPN) assigned before any characterization data exist. Detection is scored worst for the parameters that act on viral clearance, because clearance cannot be verified on the batch itself and is assured instead by control of the process (International Council for Harmonisation 2023a). Where no data or rationale is available for an assessment, the parameter is ranked at the highest level, so an absence of knowledge sends a parameter into study instead of excusing it from one.

The study type follows from that ranking. A parameter carried into the designed experiment of its step is studied multivariately, because its effect is expected to depend on the levels of the other parameters, and a parameter whose effect is expected to be independent of them (or whose range is already covered by platform experience) is studied univariately. 1 quality-linked parameter is studied univariately with a documented justification. It is the anion exchange operating flow rate, and the justification is given in RA-001 and in the plan for that step.

Table 4.1 shows where the experimental effort falls. The production bioreactor carries the largest number of parameters and the largest number of high priority ones, which follows from its forming most of the quality attributes (Section 3). The multivariate set is not the same as the set of quality-linked parameters, and more parameters enter the Protein A design than are linked to a quality attribute, because a parameter that governs the composition of the pool is studied in the same runs whether its effect falls on quality or on yield.

2 of the 8 steps carry no multivariate design at all. Harvest and clarification (Step 4) and ultrafiltration and diafiltration (Step 10) neither form nor clear a product quality attribute, so their parameters are studied univariately and are assessed against process performance and against the consistency of the material they deliver. No designed experiment is planned for either step.

The ranges taken into study are wider than the ranges the process is controlled to. For every parameter in the campaign the characterization range brackets the normal operating range, and both are tabulated in the plan for the step. Studying beyond the control range is what makes a proven acceptable range (PAR) provable at all, and it also supports later movement inside the design space (CMC Biotech Working Group 2009). The width of each range is justified parameter by parameter in the individual plans, against the capability of the equipment and against the point at which the step is known or expected to fail.

A parameter that shows no measurable effect is not dropped from the campaign. It is classified on the evidence of the study that found no effect, and the range over which no effect was found stays in the knowledge space as evidence that the attributes are robust to it (which is also what allows a later change within that range to be assessed on paper). The same holds for a parameter studied univariately.

5 Scale-down model strategy

Every study in the campaign is executed at scale-down, so qualification of the model is a precondition for the study that uses it. The conclusions of these studies are applied directly to the commercial process, and that is what makes the qualification necessary (U.S. Food and Drug Administration 2011). Qualification is performed under SOP-1001, and each plan states the qualification specific to its step.

The models are designed on the principle that the variables governing the attribute of interest are held equivalent across scales. The scale-down bioreactor is a stirred tank of the same design class as the commercial vessel, with equivalent control of pH, dissolved oxygen, temperature and nutrient addition. The scale-independent variables (pH, temperature, culture duration, seeding density) are operated at the values proposed for commercial manufacture. The scale-dependent variables (agitation, gas flow, head pressure, working volume) are set so that the small scale reproduces the performance of the full scale (CMC Biotech Working Group 2009).

The chromatography models use the same resin and the same bed height as the commercial columns, with linear velocity, load in grams per litre of resin, and all wash and elution volumes normalized to column volumes; column efficiency is verified by plate count and peak asymmetry before use. The filtration models use the same membrane and the same area-normalized load, so that flux decay and the retention mechanism are the ones the commercial filter will see. The hold steps use the same vessel geometry class and the same mixing and temperature control.

A model is qualified by running it at the target condition and comparing its performance and product quality with data from pilot and commercial batches. The comparison is statistical, and the attributes compared are the ones the model is required to predict. A model is qualified when no statistically significant difference is found for those attributes, or when a difference is found, explained by a known scale effect and shown to be conservative for the attribute at risk, in which case the limitation is recorded with the qualification. Each plan names the attributes compared, the number of replicate runs and the criterion applied.

No scale-down model reproduces every aspect of the commercial process. Gas transfer, carbon dioxide accumulation and mixing time are scale dependent in the bioreactor, and they are matched on performance and not on geometry. Where a variable cannot be matched, the plan states how the study is bounded and the report states what the study does not cover. The qualification data are held in the qualification records under SOP-1001 and are summarized in the materials and methods section of each report (they are not reproduced here, and no conclusion in this plan depends on them).

A change to a qualified model during the campaign requires re-qualification before further runs, together with an assessment of the studies already executed on the earlier configuration. Where re-qualification shows that the earlier configuration was not representative, the affected study is invalidated for the definition of an operating region and is re-executed in full (Section 7).

5.1 Analytical methods

Every response measured in the campaign is measured by a validated method. The methods and the controlled documents that govern them are listed in Table 5.1. Sampling points and sample handling are fixed in the plan for the step before execution.

Table 5.1: Controlled procedures and validated analytical methods used across the characterization campaign. Numbers are placeholders for this synthetic package.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP

Reference	Title	Type
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation

Assay variability is part of the variability these studies observe, and each report separates the two where the method validation supports the separation. Where an effect is statistically significant but small compared with the precision of the assay that measured it, the report says so and does not treat the effect as practically meaningful. Where an assay is not sensitive enough to quantify clearance across a step, clearance is determined by a spiking study on the qualified scale-down model, and the plan for that step says so (CMC Biotech Working Group 2009).

6 Common statistical approach

The multivariate studies are executed in two stages, and the two stages carry different claims. The screening design identifies which parameters have a measurable effect on each response. The response surface design is the predictive model, and it is the model from which an operating region or a design space is derived. A screening design in this campaign is near saturated and cannot resolve all the interactions between parameters, so no operating region is claimed from one (CMC Biotech Working Group 2009). This division holds for every multivariate study in the campaign, and the individual plans apply it without restating it.

The planned structure of those studies is given in Table 6.1. Each screening design is a two-level factorial or fractional factorial in the screening factors of the step, augmented with centre points at the target condition, and each response surface design is a face-centred central composite in the factors that screening carries forward, again with centre points. Run order is randomized. The design family, its resolution and its full run list are given in the plan for the step (the planned matrices are tabulated in its appendices, in coded and in natural units).

Table 6.1: Planned structure of the multivariate characterization studies. Run counts include the centre points of each design.

Step	Unit operation	Screening factors	Screening runs	Response-surface runs	Responses measured
3	Production Bioreactor	5	19	28	5
5	Protein A Chromatography	4	19	28	3
6	Low-pH Viral Inactivation	3	11	18	3
7	Cation Exchange Chromatography	4	19	28	3
8	Anion Exchange Chromatography	4	19	28	4
9	Small-Virus Retentive Filtration	2	7	12	3

The run counts in Table 6.1 are planned counts. The factor set of a response surface design is fixed only once the screening study of that step has been analysed, so a screening result that carries more factors forward than expected requires more runs, and the plan for the step is revised before the response surface study is executed.

Centre points serve two purposes in every study. They estimate pure error independently of the fitted model, which is what makes a lack-of-fit test possible, and they measure the reproducibility of the step at its target condition, which is the first result reported from any of these studies.

Effects are estimated on coded factors, where the low and high edges of the characterization range are -1 and $+1$ and the set-point is 0. Coding puts every parameter on the same scale, so an effect is the change in a response over half the characterization range and effects are comparable between parameters measured in different units. Terms are retained at $p < 0.05$. The response surface model is a full quadratic in the factors carried forward (main effects, two-factor interactions and quadratic terms), and it is reported with its coefficients, their standard errors and their p-values.

Model adequacy is assessed before a model is used for anything else. Each fitted response surface is tested for lack of fit against the pure error of the centre points, and its residuals are examined against the predicted values and on a normal quantile plot. The coefficient of determination is reported, and it is read together with the lack-of-fit test and not on its own. A model that fails any of these checks is not used to define an operating region: the effects that screening established are still reported, and the step is then bounded by univariate work and by prior knowledge.

Two ranges are derived from each adequate response surface model, and they answer different questions. The first is the range of a parameter over which the attribute stays inside its acceptance criterion with the other parameters of the step held at their set-points. The second is the range that survives when the other parameters vary within their normal operating ranges at the same time, which is the range the step can be operated over while the rest of the step varies as it normally does, and it is the one carried into the control strategy.

A design space is claimed for a step only where the response surface models of that step define a region in which every attribute the step governs stays inside its acceptance criterion simultaneously (International Council for Harmonisation 2009). The region is bounded by the characterization ranges that produced it, and the worst-case corner of the region (the combination of factor levels at which the governing attribute is closest to its limit) is identified and shown to satisfy the criteria. A response surface predicts mean levels, so a region defined on the mean gives roughly even odds at its own edge (CMC Biotech Working Group 2009). That is why the control strategy is built on the capability estimate and not on the region alone.

Capability is estimated at the level of the drug substance and not step by step. The linked process model is run for 2,000 simulated commercial batches, with every parameter of every step drawn independently around its set-point at a spread set by the width of its normal operating range and truncated at the characterization range, and the simulated distribution of each attribute is then compared with its acceptance criterion. The comparison is one-sided for the impurities and for the viral clearance attributes, because only one limit is meaningful for them, and two-sided for the attributes that carry a target range.

The capability estimate inherits the bounds of the models it is built from. It assumes that the response surfaces hold over the ranges simulated, that the scale-down models represent the commercial process, and that the parameters vary independently of one another. It is reported per attribute in PCMR-001 and is confirmed at scale by the Stage 2 qualification batches.

7 Acceptance criteria framework

Four decisions are made in every characterization study, and this section defines what each of them requires. They are whether the scale-down model is qualified, whether a fitted model is adequate, whether an operating region is acceptable, and how each parameter is classified. The numeric limits belong to the individual plan, because they depend on the attributes the step governs and on the precision of the assays that measure them.

7.1 Which criterion a step is judged against

Before any of those four decisions can be made, the criterion itself has to be fixed, and for most steps it is not the drug substance specification. The specifications in Table 3.1 apply to the drug substance. An intermediate pool is not the drug substance, and judging one against those specifications gives an answer that is wrong in both directions. The capture pool carries about 9,125 ng/mg host cell protein against a drug substance limit of 100 ng/mg, so the comparison condemns a step that is performing normally. At the last step the same comparison is too generous, because it leaves no margin for the variation of a batch that has nowhere left to be cleaned.

Each step is therefore given an in-process limit, and the campaign uses one of three procedures to set it. Which procedure applies depends on what the downstream train does to the attribute, not on the preference of the author.

The first procedure is backward calculation from the drug substance specification. It is the default, and it applies to every attribute that a later step still clears. The specification is carried back through the clearance that the downstream steps deliver in the nominal train, which gives the concentration at this step from which the rest of the process can still reach the specification. That concentration is then divided by an assurance margin. The margin is not decoration. A limit set at the undivided ceiling would leave the drug substance dependent on every downstream step delivering its nominal clearance exactly, which is a break-even point and not a control.

Host cell protein at the capture step is the worked example. The drug substance limit is 100 ng/mg. Cation exchange clears a factor of 78.0 and anion exchange a further 6.2, so a capture pool at 48,260 ng/mg would still reach the specification if both downstream steps performed exactly as they do in the nominal train. Dividing by an assurance margin of 4.5 gives the in-process limit of 10,724 ng/mg that PCP-005 and PCR-005 apply. The margin is chosen so that the limit also sits above the nominal pool and above the design centre of the study, because a limit below either would put the process outside its own limit while running at its set-point.

The second procedure is a capability alert limit. Where no later step modifies the attribute, there is no clearance to calculate back through. The drug substance specification already applies at the step that forms the attribute, and for the glycans and the charge variants it is far wider than the spread the process itself shows, so it would not bound the step at all. The limit is then set from demonstrated capability, at the mean plus 2.5 standard deviations of the commercial-scale simulation. This is an alert limit and not a specification. A batch beyond it is investigated, not rejected.

The third procedure is the modular clearance claim. A viral clearance step is judged on the log reduction claimed for it, less an allowance of 0.35 $\log_{10}$ for the variability of the spiked clearance assay. Where the claim itself is credited from this process's own studies rather than from platform data, the step-level requirement is instead the cumulative requirement less the clearance credited to the other orthogonal steps, which is the modular calculation described in Section 6.

12 in-process limits are set across the campaign by these three procedures. None of them is a typed number. Each is a rule evaluated against the seeded process model, so a change to the process or to a clearance factor moves the limits with it, and the plans and reports for the individual steps state the value each rule produces together with the procedure that produced it. Two consequences follow and are stated here because they surprise readers who expect the drug substance specification everywhere. An in-process limit is tighter than the specification it derives from, so a proven acceptable range measured against it can be narrower than the range that was characterized. It follows that the design space for a step is generally smaller than its characterized region, because a parameter with a narrowed range is a parameter some of whose characterized settings are not acceptable.

A scale-down model is qualified when the performance and quality attributes it is required to predict are statistically indistinguishable from the commercial-scale data at the target condition. A difference does not by itself disqualify a model: where it is explained by a known scale effect and is conservative for the attribute at risk, the model is qualified with that limitation recorded, and the report states which conclusions the limitation touches. A model that is not qualified cannot support an operating range, and the study that used it is repeated.

A fitted model is adequate when it has significant terms at $p < 0.05$, no significant lack of fit against pure error, and residuals consistent with the assumptions of the fit. All three are required, and the criteria for each (the significance level, the lack-of-fit p-value and the residual checks) are stated in the plan for the step. A response with no significant terms is reported as such, because a null result bounds how much of the observed variation the parameters studied can account for.

An operating region is acceptable when three conditions hold together. Every attribute the step governs is predicted to stay inside the in-process limit set for it under Section 7.1 across the region, that prediction holds with the other parameters of the step varying within their normal operating ranges, and the worst-case corner of the region (identified from the signs of the model coefficients) also satisfies those limits. Where the worst corner does not, the region is the part of the characterized space that does, and the report states how much of the characterized space that is. A region defined this way is claimed only inside the characterization ranges that produced it, and it remains a statement about mean levels until the capability analysis converts it into a statement about batches.

Capability is estimated as described in Section 6. This plan does not set one minimum capability index for the whole campaign. An acceptable value depends on the criticality of the attribute and on how much of the observed variation belongs to its assay, so each plan states the minimum index its step's attributes must meet together with the reason for the value chosen. Where an attribute is governed by more than one step, the criterion is applied to the cumulative position at the drug substance and not to the individual steps, and it is assessed in PCMR-001.

Parameter classification is performed under SOP-4001 and uses the same four classes throughout the campaign. A parameter whose variation affects a critical quality attribute is critical; where such a parameter is easy to control and the risk of leaving the design space is low it is classified as a well-controlled critical process parameter (WC-CPP), and where that risk is not low it is classified as a critical process parameter (CPP). A parameter that affects process performance or consistency but not product quality is a key process parameter (KPP), and a parameter with no demonstrated effect on either is a general process parameter (GPP). The class depends on the control capability of the receiving site as well as on the study result, so it is assigned after the study and recorded in the report for the step.

A parameter that shows no effect is still classified, and its classification is supported by the range over which no effect was seen. Nothing is left unclassified at the end of the campaign.

A study is valid when it is executed within its approved protocol. Excursions are recorded as deviations, assessed for their effect on the conclusions and described in the report for the step, whether or not they changed the outcome. Where a deviation makes the material or the execution unrepresentative of the commercial process, the affected design is invalidated for the definition of an operating region and is re-executed in full on representative material. The invalidated data are retained and are referenced in the report as evidence of the root cause. They are not used to define ranges and are not analysed as if they were valid.

8 Register of characterization plans

Each unit operation is planned in its own protocol, and each protocol is approved before the first study run of its step. The protocols are listed in Table 8.1 with the unit operation each covers and the report it pairs with. Every protocol follows the structure and the methods defined in this plan and adds the design, the ranges and the acceptance criteria of its own step.

Table 8.1: Register of the per-unit-operation characterization plans and their paired reports.

Step	Unit operation	Characterization plan	Paired report
3	Production Bioreactor	PCP-003	PCR-003
4	Harvest and Clarification	PCP-004	PCR-004
5	Protein A Chromatography	PCP-005	PCR-005
6	Low-pH Viral Inactivation	PCP-006	PCR-006
7	Cation Exchange Chromatography	PCP-007	PCR-007
8	Anion Exchange Chromatography	PCP-008	PCR-008
9	Small-Virus Retentive Filtration	PCP-009	PCR-009
10	Ultrafiltration / Diafiltration	PCP-010	PCR-010

9 Deliverables and schedule

The deliverables of the campaign are listed in Table 9.1. The sequence is fixed by dependency and not by date. A characterization plan is approved before the studies it plans are executed, a characterization report is issued once the studies of its step have been analysed, and the master report is issued after the last of those reports. Dates for each deliverable are controlled in the project schedule and are outside this plan.

Table 9.1: Deliverables of the characterization campaign and their position in the sequence.

Deliverable	Document	Position in the sequence
Process Transfer Plan	PTP-001	Approved before this plan
Pre-Characterization Process Risk Assessment	RA-001	Approved before this plan
Process Characterization Master Plan	PCMP-001	This document
Process Characterization Plan (one per unit operation)	PCP-003 to PCP-010	Approved before the first study run of its step
Process Characterization Report (one per unit operation)	PCR-003 to PCR-010	Issued once the studies of its step have been analysed
Process Characterization Master Report	PCMR-001	Issued after the last characterization report

The campaign is complete when PCMR-001 is approved. That report is the input to the Stage 2 qualification protocol and to the process sections of the licence application.

10 Approvals

This plan is approved by the functions listed in Table 1. A change to the framework defined here is made by revising this plan, and the individual plans that depend on the changed element are revised with it. Changes are assessed under the pharmaceutical quality system (International Council for Harmonisation 2008).

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2008. *ICH Q10: Pharmaceutical Quality System*. ICH.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.